OPTIMAL CONTRACT AND GENERAL CARBON THEORY FOR FLEXIBLE COMPUTING LOADS: EXISTENCE AND UNIQUENESS, A UNIFORM HOMOTOPY CONDITION, THE GENERAL-F EMISSION KEYSTONE, CONNECTEDNESS, AND SHARPNESS

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

ABSTRACT. This is the Role 1 Week 2 deliverable. It hardens the Week 1 carbon theory (`role1_carbon_theory.tex`) into five results—four proved, one resolved numerically—taking the same model and notation and staying in Role 1's large-FCL mandate (the one small-FCL touch is flagged as a conjecture pending Role 4). *One*, the regulated program has a maximizer, it is unique, and the within-period dispatch $d^*(x) = \min\{(K_F - x)_+, K_M\}$ is the unique optimal rule; uniqueness comes not from global concavity (the profit is submodular in (K_R, K_F), so it fails) but from a *recursive-triangular* structure—the distribution-free cutoffs pin K_F and then K_M as unique functions of K_R, after which a single scalar diagonal-dominance (reduced second-order) condition pins K_R. *Two*, support-wide monotonicity of the shift ratio $g = F_{K_R}/f$ makes the realized two-point endpoint inequality hold *uniformly* along the whole C_M-homotopy, even as the dispatch window moves—the exact content the Week 1 "integrate-downward" step needs. *Three* (the keystone), substituting the equilibrium map $K^*(C_R)$ into the exact inequality (C) yields the endogenous carbon condition, and we give a *primitive* sufficient condition—the endpoint shift-ratio condition plus a two-zone density (tail/skew) bound—that would make carbon fall at a *finite interior* optimum. We then settle whether it ever fires, with two *proved* results and a numerical benchmark. First, a fossil floor: $\mathbb{E}[\text{fossil}^*] \geq K_F^*(R - c - C_M)/(R - c)$ (and $\geq K_M^* - K_R^*$), since an FCL exceeding the maximal renewable surplus $\leq K_R^*$ cannot be fully surplus-powered. Second—the key structural point—we prove the carbon sign is *irreducibly distributional*: holding the equilibrium capacities and cutoffs fixed, the admissible conditional laws make Δ range over an interval that straddles zero, so capacities and cutoffs together cannot determine $\operatorname{sign}\Delta$ and Week-1's conjecture admits no proof from the distribution-free structure alone. The residual question is then the sign of a cutoff-computable benchmark Δ_0 plus a density tilt; numerically, at every large-FCL interior optimum scanned $\Delta_0 < 0$ (≤ -1.17) and the tilt reinforces it, $\Delta < 0$ (≤ -1.4). Carbon strictly falls only in the cheap-renewable corner $K_R^* \to \infty$. So the honest reading is *negative*: interior carbon reduction is a corner phenomenon (the small-FCL regime is Role 4's, not claimed here). *Four*, the carbon-reduction set $\{C_R : \Delta(C_R) \geq 0\}$ is bounded above and non-empty—its non-emptiness coming from that power-law corner, not from an interior optimum—and is a single interval under an explicit monotone-renewable-shadow-value condition that, we note, already encodes the single-crossing property it certifies. *Five*, the sharp necessary-and-sufficient condition for $K_R^* > K_R^\circ$ is the *path integral* of the realized endpoint gap; pointwise-monotone g is sufficient but not necessary; and the two-sided bounds on the carbon saving are tight, attained by explicit degenerate conditional laws. Every identity is checked against quadrature of the profit integral in `verify_week2.py`.

1. SETUP

We keep the model and notation of `role1_carbon_theory.tex`. A regulated utility chooses renewable capacity K_R, dispatchable fossil K_F, and flexible-computing-load (FCL) capacity K_M. Residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f(x; K_R) > 0$ on $(-K_R, \bar{D})$, and $F_{K_R} := \partial F/\partial K_R \geq 0$, extended by $F \equiv 0$ below $-K_R$ and $F \equiv 1$ above $\bar{D}$. Per-MWh: FCL revenue R, fossil cost c, price p, blackout penalty β; amortized capital costs C_R, C_F, C_M.

Date: Working results, Week 2.

Assumption 1 (Regularity, as in Week 1)**.** $R > c + C_M$, $\quad R < c + C_F + C_M$, $\quad \beta + p > c + C_F$, $\beta + p > R$.

Under the dispatch rule $d(x) = \min\{(K_F - x)_+, K_M\}$ the unified profit is

$$\Pi = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F\,dx - c\int_{-K_M}^{K_F-K_M}(1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \quad (1)$$

with gradient (Week 1, Lemma 1)

$$g_M := \Pi_{K_M} = (R-c)F(K_F - K_M) + cF(-K_M) - C_M, \quad (2)$$

$$g_F := \Pi_{K_F} = R[F(K_F) - F(K_F - K_M)] - c[1 - F(K_F - K_M)] + (\beta + p)[1 - F(K_F)] - C_F, \quad (3)$$

$$g_R := \Pi_{K_R} = R\int_{K_F-K_M}^{K_F} F_{K_R}\,dx + c\int_{-K_M}^{K_F-K_M} F_{K_R}\,dx + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R. \quad (4)$$

We work in the *large-FCL interior* regime $K_M \geq K_R$ (so $F(-K_M) = f(-K_M) = F_{K_R}(-K_M) = 0$), the regime in which the Week-1 carbon theory lives; throughout, "interior" means $0 < K_R < \infty$ and the cutoffs below sit strictly inside $(0, 1)$. Write $a := K_F - K_M$ for the lower dispatch endpoint and recall the distribution-free cutoff *values* (Week 1, Lemma 2)

$$\varphi_1 := F(a; K_R) = \frac{C_M}{R-c}, \qquad \varphi_2 := F(K_F; K_R) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \qquad \varphi^\circ := F(K_F^\circ; K_R^\circ) = \frac{\beta + p - c - C_I}{\beta + p - c} \quad (5)$$

Under Assumption 1, $0 < \varphi_1 < 1$ and $0 < \varphi_2 < 1$; the interior FCL (adoption) condition $K_M > 0$ is $\varphi_2 > \varphi_1$, equivalently $C_M < \bar{C}_M := (R-c)\varphi^\circ$.

2. Existence, uniqueness, and contract optimality

The optimum is not delivered by concavity. By the Week-1 cross-partial $\Pi_{K_R K_F} = (R - \beta + p)F_{K_R}(K_F) - (R-c)F_{K_R}(a) \leq 0$, Π is *submodular* in (K_R, K_F), so the Hessian is not globally negative semidefinite and global concavity is unavailable. Uniqueness instead follows from a recursive-triangular structure together with a single scalar condition.

2.1. Dispatch optimality.

Lemma 1 (Unique optimal dispatch via strict $R > c$)**.** *Fix capacities (K_R, K_F, K_M) and a residual-demand realization x. Let the within-period payoff from serving s units of consumer demand and dispatching d units of FCL be*

$$\pi_x(s,d) = p\,s - c\,(s+d) + R\,d - \beta\,((x)_+ - s), \qquad 0 \leq s \leq \min\{(x)_+, K_F\},\ \ 0 \leq d \leq K_M,\ \ s+d \leq K_F.$$

Under Assumption 1 the unique maximizer is $s^* = \min\{(x)_+, K_F\}$ and $d^*(x) = \min\{(K_F - x)_+, K_M\}$.

Proof. π_x is affine in (s,d) with $\partial\pi_x/\partial s = \beta + p - c > 0$ and $\partial\pi_x/\partial d = R - c > 0$, both strict under Assumption 1 ($\beta + p > c + C_F \geq c$ and $R > c + C_M > c$). A strictly increasing affine objective on a compact polytope is maximized at the unique vertex reached by pushing both coordinates up to their constraints: first $s^* = \min\{(x)_+, K_F\}$ (serving consumers dominates because $\beta + p - c > R - c$, so consumer service takes priority for fossil headroom), then d up to the remaining headroom, $d^* = \min\{K_M, K_F - s^*\} = \min\{(K_F - x)_+, K_M\}$. Strictness of both slopes makes the maximizer unique; at $R = c$ the d-slope vanishes and any $d \in [0, \min\{(K_F - x)_+, K_M\}]$ would be optimal. Hence strict $R > c$ is exactly what pins d^*.[1] □

[1] Verified in `verify_week2.py` T1a: a grid search of π_x over (s,d) returns (s^*, d^*) at every sampled x.

2.2. **Existence.**

Proposition 1 (A maximizer exists)**.** Π *is continuous on* $\mathbb{R}^3_+$*, and there is a box* $[0,\bar{K}]^3$ *that no maximizing sequence leaves. Hence the set of maximizers is nonempty and compact.*

Proof. Continuity is immediate: each integrand in (1) is bounded and $F(\cdot;\cdot)$ is jointly continuous. We do not claim coercivity along all rays; we show each *own* partial turns strictly negative once its own capacity is large. As $K_M \to \infty$ (others fixed), $K_F - K_M \to -\infty$, so $F(K_F - K_M) \to 0$ and $g_M \to -C_M < 0$. As $K_F \to \infty$, $F(K_F - K_M) \to 1$ and $g_F \to -C_F < 0$. As $K_R \to \infty$, $F_{K_R}(x; K_R) \to 0$ pointwise and, for the bounded-density families we use, $\int F_{K_R}\, dx \to 0$ by dominated convergence ($F_{K_R} = \mathbb{E}[\tilde{S} f_{\tilde{D}}]$ is dominated by $\sup f_{\tilde{D}} < \infty$), so $g_R \to -C_R < 0$. Thus there is $\bar{K} < \infty$ with $\Pi_{K_i} < 0$ whenever $K_i > \bar{K}$ (others in $[0,\bar{K}]$): on each face $K_i = \bar{K}$ the gradient points strictly *inward*, so Π has no larger value outside $[0,\bar{K}]^3$ than on its boundary, and no maximizing sequence escapes the box. By Weierstrass on the compact box a maximizer exists.[2] □

2.3. **Uniqueness by recursive triangularity.** Fix K_R. By (5) the equations $g_M = 0,\ g_F = 0$ are equivalent to $F(a; K_R) = \varphi_1$ and $F(K_F; K_R) = \varphi_2$, two *distribution-free* targets. Since $x \mapsto F(x; K_R)$ is a strictly increasing bijection from $(-K_R, \bar{D})$ onto $(0,1)$, each target has a unique preimage:

$$K_F(K_R) = F^{-1}(\varphi_2; K_R), \qquad a(K_R) = F^{-1}(\varphi_1; K_R), \qquad K_M(K_R) = K_F(K_R) - a(K_R) > 0, \tag{6}$$

all C^1 in K_R (implicit function theorem, $\partial F/\partial x = f > 0$). This is the triangle: levels one and two (K_F, K_M) are pinned by K_R with no distributional input. Substituting (6) into g_R defines the reduced renewable condition

$$\Psi(K_R) := g_R(K_R, K_F(K_R), K_M(K_R)), \qquad \Psi(K_R^*) = 0. \tag{7}$$

Lemma 2 (The (K_F, K_M) block is negative definite given interiority)**.** *At any large-*FCL *interior point (so* $f(a), f(K_F^*) > 0$*, the cutoffs strictly inside the support) the* 2×2 *Hessian block in* (K_F, K_M)*,* $\mathcal{M} = \begin{pmatrix} \Pi_{K_F K_F} & \Pi_{K_F K_M} \\ \Pi_{K_F K_M} & \Pi_{K_M K_M} \end{pmatrix}$*, satisfies* $\Pi_{K_F K_F} < 0$*,* $\Pi_{K_M K_M} < 0$*, and*

$$\det \mathcal{M} = (R-c) f(a)\, (\beta + p - R) f(K_F) \ > \ 0,$$

so $\mathcal{M} \prec 0$ *under Assumption 1 given interiority (no* distributional *condition beyond* $f > 0$ *at the cutoffs).*

Proof. From (2)–(3) in the large-FCL regime, $\Pi_{K_M K_M} = -(R-c) f(a) < 0$, $\Pi_{K_F K_M} = (R-c) f(a)$, and $\Pi_{K_F K_F} = (R - \beta + p) f(K_F) - (R-c) f(a) < 0$ (both terms negative since $R < \beta + p$, $R > c$). Then $\det \mathcal{M} = \Pi_{K_F K_F} \Pi_{K_M K_M} - \Pi^2_{K_F K_M} = [-(R-c) f(a)]\, (R - \beta + p) f(K_F) + (R-c)^2 f(a)^2 - (R-c)^2 f(a)^2 = (R-c) f(a)(\beta + p - R) f(K_F) > 0$. Negative diagonal and positive determinant give $\mathcal{M} \prec 0$.[3] □

Condition 1 (Reduced second-order / diagonal dominance)**.** Along the interior locus (6), the Schur complement of the renewable coordinate is negative,

$$\Psi'(K_R) = \Pi_{K_R K_R} - \begin{pmatrix} \Pi_{K_R K_F} & \Pi_{K_R K_M} \end{pmatrix} \mathcal{M}^{-1} \begin{pmatrix} \Pi_{K_R K_F} \\ \Pi_{K_R K_M} \end{pmatrix} < 0. \tag{8}$$

Condition 1 is a *single scalar* inequality—the second-order condition of the one-dimensional reduced problem $\Psi(K_R) = 0$, i.e. negative diagonal dominance of the Hessian in the renewable row after the (K_F, K_M)-block has been eliminated. It is strictly weaker than global concavity (which submodularity rules out) and it is exactly what lapses, becoming an equality, at the power-law knife-edge where $\Psi \equiv$ const. *Primitive content.* $\Psi(K_R) + C_R$ is the marginal value of renewables

[2] T1b: the own partial is strictly negative at $K_i = 60$ with the others moderate.

[3] T1c: $\det \mathcal{M}$ matches the closed form and $\operatorname{eig} \mathcal{M} < 0$ at the solved optimum, on Beta(2, 2) and Beta(2, 3) scale mixtures.

along the re-optimization locus; $\Psi' < 0$ says this marginal value is strictly decreasing in K_R—the renewable problem is single-peaked along the locus. The power-law family is the exact boundary ($\Psi' \equiv 0$, constant marginal value: the knife-edge of Week-1, Prop. knife), so any family with strictly diminishing renewable returns along the locus—which we verify for the Beta scale mixtures by the eigenvalue scan of §7—satisfies it strictly. (We give a checkable scalar, not a closed-form primitive on F; deriving $\Psi' < 0$ from a density condition on F in general is open.)

Theorem 1 (Existence, uniqueness, optimality)**.** *Under Assumption 1 and Condition 1, the profit* Π *has* at most one *interior critical point; it exists whenever the program's optimum is interior (the cutoffs* (5) *lie strictly in* $(0,1)$ *with* $K_M > 0$*—the Week-1 large-*FCL *adoption regime,* $C_M < \bar{C}_M$ *and* C_R *not so large that* $K_R^* = 0$*), in which case it is the* unique *interior maximizer* (K_R^*, K_F^*, K_M^*)*, the Hessian is negative definite there, and* $d^*(x) = \min\{(K_F - x)_+, K_M\}$ *is the unique optimal dispatch.*

Proof. Any interior critical point solves $\nabla\Pi = 0$, hence (cutoff Lemma) lies on the locus (6) and satisfies $\Psi(K_R) = 0$. By (7) and the chain rule, $\Psi'(K_R)$ equals the Schur complement (8) (the total derivative of g_R along the locus, after eliminating (K_F, K_M) through $g_M = g_F = 0$). Lemma 2 gives $\mathcal{M} \prec 0$, so $\mathcal{M}^{-1} \prec 0$ and the quadratic form in (8) is ≥ 0; Condition 1 makes $\Psi'(K_R) < 0$, so Ψ is strictly decreasing and has *at most one* zero—hence at most one interior critical point. When the optimum is interior (the stated regime), a maximizer exists by coercivity (Proposition 1) and lies in the interior, so it is a critical point and therefore *the* unique one K_R^*; (6) returns unique K_F^*, K_M^*. By the Schur determinant identity $\det \mathcal{H} = \det \mathcal{M} \cdot \Psi'(K_R)$ with $\det \mathcal{M} > 0$ and $\Psi' < 0$, the leading-principal-minor signs alternate, so $\mathcal{H} \prec 0$ at the optimum; the dispatch claim is Lemma 1.[4] □

Remark 1. The triangle is the whole point. Concavity would ask the full 3×3 Hessian to be negative definite as a joint hypothesis; the recursive structure instead *derives* negative definiteness from two cheap facts—an interiority-only 2×2 block (Lemma 2) and one scalar slope (Condition 1)—which is why the submodularity in (K_R, K_F) is no obstruction.

Corollary 1 (The equilibrium map is C^1)**.** *On the open large-*FCL *interior region where Condition 1 holds, the optimum* (K_R^*, K_F^*, K_M^*) *is a* C^1 *function of the parameters* (C_R, C_M)*; likewise the no-*FCL *optimum* (K_R°, K_F°)*. Consequently* $\mathbb{E}[\textit{fossil}^*]$*,* $\mathbb{E}[\textit{fossil}^\circ]$*, and* $\Delta = \mathbb{E}[\textit{fossil}^\circ] - \mathbb{E}[\textit{fossil}^*]$ *are* C^1 *in* C_R *there.*

Proof. By Theorem 1, $\mathcal{H} \prec 0$ at the optimum, so the Jacobian of the C^1 map $K \mapsto \nabla\Pi(K; C_R, C_M)$ in K is nonsingular; the implicit function theorem gives $(C_R, C_M) \mapsto K_R^*, K_F^*, K_M^*$ in C^1. The no-FCL system $g_F = g_R = 0$ ($K_M \equiv 0$) has Jacobian $\mathcal{M}$'s no-FCL analogue, nonsingular by the same block argument, giving $K_R^\circ, K_F^\circ \in C^1$. Each fossil expectation is a smooth functional of the capacities, so $\Delta \in C^1$ by the chain rule.[5] □

3. The shift-ratio condition, uniform along the homotopy

Week-1 Theorem 1 signs $K_R^* > K_R^\circ$ by integrating $dK_R^*/dC_M \leq 0$ downward from $\bar{C}_M$. The cofactor sign it integrates is

$$\operatorname{cof}_{K_M K_R}(\mathcal{H}) \propto f(K_F^*) f(a) \big[g(a) - g(K_F^*)\big], \qquad g(x; K_R) := \frac{F_{K_R}(x; K_R)}{f(x; K_R)}, \quad a = K_F^* - K_M^*,$$

[4] T1c: $\Psi(K_R)$ is strictly decreasing on a scan of the locus with its unique zero at the solved K_R^* ($|\Psi(K_R^*)| < 10^{-6}$), and $\operatorname{eig} \mathcal{H} < 0$.

[5] Smoothness of $K_R^*(\cdot)$ is exhibited numerically by the T5a fundamental-theorem-of-calculus test (its finite-difference derivative integrates back to the level on the interior band). The only non-smooth point is the large-/small-FCL crossing, which is outside this regime (Role 4).

so the step needs the *realized two-point endpoint inequality* $g(a) \geq g(K_F^*)$ to hold at *every* C_M on the path—even though the window $(a, K_F^*) = (K_F^* - K_M^*, K_F^*)$ moves with C_M. The following lemma supplies exactly that from a primitive, C_M-free hypothesis.

Condition 2 (Support-wide monotone shift ratio)**.** For each K_R in the equilibrium range, $x \mapsto g(x; K_R)$ is non-increasing on the support $(-K_R, \bar{D})$.

Lemma 3 (Uniform endpoint inequality along the homotopy)**.** *Suppose Condition 2 holds. Then for every $C_M \in (0, \bar{C}_M)$, at the interior optimum the window endpoints satisfy $-K_R^* < a(C_M) < K_F^*(C_M) < \bar{D}$ and*

$$g(a(C_M); K_R^*(C_M)) \;\geq\; g(K_F^*(C_M); K_R^*(C_M)), \tag{9}$$

with strict inequality wherever $g(\cdot; K_R^*)$ is strictly decreasing. Hence $\mathrm{cof}_{K_M K_R}(\mathcal{H}) \geq 0$ and $dK_R^*/dC_M \leq 0$ uniformly on the path, and the Week-1 downward integration is justified by Condition 2 alone.

Proof. At an interior optimum with $C_M < \bar{C}_M$ the adoption margin is strict, $K_M^* > 0$, so $a = K_F^* - K_M^* < K_F^*$. The cutoffs (5) give $F(a; K_R^*) = \varphi_1 \in (0,1)$ and $F(K_F^*; K_R^*) = \varphi_2 \in (0,1)$, so both endpoints lie strictly inside the support $(-K_R^*, \bar{D})$ (a point where $F \in (0,1)$ cannot be at an extended-CDF plateau). Condition 2 applied at $K_R = K_R^*$ to the two ordered interior points $a < K_F^*$ yields $g(a; K_R^*) \geq g(K_F^*; K_R^*)$, which is (9). This holds for every C_M because the hypothesis is on the whole support, independent of where the window sits; as C_M sweeps $(0, \bar{C}_M)$ the window sweeps a sub-family of ordered interior pairs, all covered at once. The cofactor and derivative signs then follow from the Week-1 cofactor identity.[6] □

Remark 2. Condition 2 is the honest primitive behind Week-1 Condition CS, which was stated on the moving window "at each point of the equilibrium path." Lemma 3 shows the moving-window statement is *implied* by the static support-wide one, so the comparative static rests on a hypothesis one can check on F without first solving for the path. The power-law family has $g(x) = (\bar{D} - x)/(\bar{D} + K_R)$, affine decreasing for every α, hence satisfies Condition 2 trivially.

4. The emission condition for general F (keystone)

Week 1 closed the carbon sign only on the power-law/uniform knife-edge (a *corner* $K_R^* \in \{0, \infty\}$) and left a non-empty strict-fall window for general F *conjectural*. We close it at a *finite interior* optimum. Recall (Week 1, Prop. 4) the exact net fossil saving $\Delta := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$ with

$$\mathbb{E}[\text{fossil}^*] = K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*), \qquad \mathbb{E}[\text{fossil}^\circ] = K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\,\mu_{\text{mid}}^\circ, \tag{10}$$

where $a = K_F^* - K_M^*$, $q^\circ = F(0; K_R^\circ)$, and $\mu_{\text{low}}^* = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (-K_R^*, a)]$, $\mu_{\text{mid}}^\circ = \mathbb{E}[\tilde{x} \mid \tilde{x} \in (0, K_F^\circ)]$. Carbon falls iff $\Delta > 0$, i.e. inequality (C) of Week 1 holds.

The endogenous condition. Substituting the equilibrium map $K_R^*(C_R), K_F^*(C_R), K_M^*(C_R)$ and $K_R^\circ(C_R), K_F^\circ(C_R)$ into (10) turns Δ into a function of primitives and C_R alone:

$$\Delta(C_R) = \Big[K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ\Big] - \Big[K_F^* - \tfrac{C_M}{R-c}(a - \mu_{\text{low}}^*)\Big], \tag{11}$$

all objects evaluated at the equilibrium. The difficulty is that the conditional means $\mu_{\text{low}}^*, \mu_{\text{mid}}^\circ$ and the capacities move with the shape of F.

Condition 3 (Two-zone density / tail-skew bound)**.** At the equilibrium, (a) $f(\cdot; K_R^*)$ is non-increasing on the FCL zone $(-K_R^*, a)$, and (b) $f(\cdot; K_R^\circ)$ is non-decreasing on the no-FCL zone $(0, K_F^\circ)$.

[6] T2: along a 12-point homotopy the realized gap $g(a) - g(K_F^*) \geq 0$ and $dK_R^*/dC_M \leq 0$ at every node; and $g(\cdot; K_R)$ is non-increasing on the full support for every sampled K_R.

Part (a) is a left-tail/skew bound on *surplus* states (deep-surplus mass not too thin); part (b) a skew bound on the no-FCL mid zone. Both are monotone-density conditions on the model primitive f (the zone endpoints are equilibrium-determined, but one obtains a fully primitive sufficient version by requiring the monotonicity on the maximal zones swept as C_M, C_R range over the adoption region). Like Condition 2, these are conditions on F, not on the solved optimum.

Theorem 2 (General-F emission keystone)**.** *Suppose the program has a finite interior optimum (Theorem 1), Condition 2 holds (so $K_R^* > K_R^\circ$ by Lemma 3), and Condition 3 holds. Then*

$$\Delta \;\geq\; \Delta_{\mathrm{LB}} := K_F^\circ\Big[(1-\varphi^\circ) + \tfrac{\varphi^\circ - q^\circ}{2}\Big] + \frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2} - K_F^*, \tag{12}$$

and therefore carbon strictly falls ($\Delta > 0$) whenever $\Delta_{\mathrm{LB}} > 0$. The bound is expressed in the equilibrium capacities and the distribution-free reliability φ° only. Caveat: *the condition $\Delta_{\mathrm{LB}} > 0$ is a property of a capacity tuple, not a guarantee that any optimum attains it; Proposition 3 reports that it is essentially never attained at a solved interior optimum, so the value of* (12) *is its* form *(it pinpoints, via Lemma 4, the floor that defeats it), not its activation.*

Proof. Under Condition 3(a), a non-increasing density on $(-K_R^*, a)$ has conditional mean at most the interval midpoint, $\mu_{\mathrm{low}}^* \leq (a - K_R^*)/2$ (the standard fact: a density non-increasing on $[\ell, u]$ has mean $\leq (\ell+u)/2$, with equality for the uniform); since the coefficient $-C_M/(R-c)$ on μ_{low}^* in (10) is negative, $\mathbb{E}[\text{fossil}^*] \leq K_F^* - \frac{C_M}{R-c}(a - \frac{a-K_R^*}{2}) = K_F^* - \frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2}$. Under Condition 3(b), a non-decreasing density on $(0, K_F^\circ)$ has conditional mean at least the midpoint, $\mu_{\mathrm{mid}}^\circ \geq K_F^\circ/2$, and since $\varphi^\circ - q^\circ > 0$, $\mathbb{E}[\text{fossil}^\circ] \geq K_F^\circ(1-\varphi^\circ) + (\varphi^\circ - q^\circ)K_F^\circ/2$. Subtracting gives $\Delta \geq \Delta_{\mathrm{LB}}$. The capacity channel $K_R^* > K_R^\circ$ (Lemma 3) is what makes the renewable term $\frac{C_M}{R-c}\cdot\frac{a+K_R^*}{2}$ large: a strictly larger K_R^* widens the FCL surplus zone and lowers $\mathbb{E}[\text{fossil}^*]$. Hence the listed hypotheses—endpoint shift-ratio (channel) plus the two-zone bound—are jointly sufficient for $\Delta > 0$.[7] □

Whether the bound is *activated* is the real question. We resolve it with two proved facts—a fossil floor and an exact impossibility—plus a numerical benchmark. We stay in the large-FCL regime throughout (Role 1's mandate); the small-FCL regime is Role 4's and we make no claim about it.

Lemma 4 (Fossil floor)**.** *In the large-FCL regime ($K_M^* \geq K_R^*$), using only the cutoff φ_1,*

$$\mathbb{E}[\textit{fossil}^*] \;\geq\; K_F^*\,(1-\varphi_1) = K_F^*\,\frac{R-c-C_M}{R-c} \qquad \textit{and} \qquad \mathbb{E}[\textit{fossil}^*] \geq K_M^* - K_R^*. \tag{13}$$

The first holds because $F \leq F(a) = \varphi_1$ on $(-K_M^*, a)$, so $\mathbb{E}[\textit{fossil}^*] = \int_{-K_M^*}^{a}(1-F) \geq (a+K_M^*)(1-\varphi_1) = K_F^*(1-\varphi_1)$; the second because $F \equiv 0$ on $(-K_M^*, -K_R^*)$. Physically the second is the floor*: the FCL exceeds the maximal renewable surplus $\leq K_R^*$, so an $K_M^* > K_R^*$ FCL draws at least $K_M^* - K_R^*$ fossil even at deepest surplus.*[8]

The floor lower-bounds $\mathbb{E}[\text{fossil}^*]$ but does not by itself sign $\Delta = \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^*]$, because the channel $K_R^* > K_R^\circ$ pushes the other way. The next result explains why *no* distribution-free argument can sign it—the sign is irreducibly distributional.

Proposition 2 (The carbon sign is irreducibly distributional)**.** *Hold the equilibrium capacities $(K_F^*, K_M^*, K_R^*, K_F^\circ)$ and the surplus probability q° fixed. By Week-1's sufficient-statistic theorem, Δ is then* affine *in the two conditional means $\mu_{\mathrm{low}}^* \in (-K_R^*, a)$ and $\mu_{\mathrm{mid}}^\circ \in (0, K_F^\circ)$, with coefficients*

[7] T3: the sufficient bound Δ_{LB} is computed at every interior optimum scanned; the algebra $\mu_{\mathrm{low}}^* \leq (a - K_R^*)/2$, $\mu_{\mathrm{mid}}^\circ \geq K_F^\circ/2$ under (a),(b) is checked directly.

[8] The fossil formula already credits surplus absorption—a state-by-state simulation serving FCL from surplus first matches $\int_{-K_M^*}^{a}(1-F)$ to ≈ 6 digits (T3, 6.18324 vs 6.18325); the floor is the *residual* after that credit.

fixed by those capacities and cutoffs; the means themselves are not *determined by capacities and cutoffs. Writing* Δ_0 *for the value at the interval midpoints (the uniform shape),*

$$\Delta_0 := K_F^\circ\Big[(1-\varphi^\circ) + \tfrac{\varphi^\circ - q^\circ}{2}\Big] - \Big(K_F^* - \tfrac{C_M}{R-c}\cdot\tfrac{a+K_R^*}{2}\Big), \tag{14}$$

the two-zone tilts sandwich Δ *around it—*$\Delta \geq \Delta_0$ *under Cond. 3(a,b),* $\Delta \leq \Delta_0$ *under the reverse tilt (*a'*,*b'*), equality at the uniform. (Note* Δ_0 *is exactly the keystone bound* Δ_{LB} *of (12): the only "sufficient-for-fall" statement is* $\Delta_0 > 0$*, and Cond. 3 is precisely the tilt giving* $\Delta \geq \Delta_0$*.) By Week-1's tight-bounds theorem, as the means sweep their intervals* Δ *ranges over* $[\Delta_{\min}, \Delta_{\max}]$*, and at the equilibrium capacities this range* straddles zero*:* $\Delta_{\min} < 0 < \Delta_{\max}$ *(e.g. at the representative optimum* $[\Delta_{\min}, \Delta_{\max}] = [-6.43, +3.23]$*). Consequently* ***the equilibrium capacities and cutoffs together do not determine the sign of*** Δ*—a function of those quantities alone cannot equal* $\operatorname{sign}\Delta$*, since both signs occur at the same capacities and cutoffs for different admissible conditional laws. Only the conditional means break the tie, which is why Week-1's conjecture admits no proof from the distribution-free structure alone.*[9]

Proof. The affine representation and the midpoint bounds for monotone densities are Week-1's sufficient-statistic and least-favorable results, restated; reversing each monotonicity reverses the corresponding inequality, giving the sandwich. Since the two extremal tilts are both admissible at the *same* cutoffs, the cutoffs cannot decide $\operatorname{sign}\Delta$.[10] □

So the carbon question reduces to the sign of the cutoff-computable benchmark Δ_0 plus a distributional tilt. We settle both numerically.

Proposition 3 (Numerical resolution, large-FCL)**.** *Proposition 2 makes the carbon sign a genuinely distributional quantity—so its resolution* must *be numerical, and we make that resolution exhaustive. For the non-power-law Beta scale-mixture family (*$\tilde{D} \sim \bar{D}\,\mathrm{Beta}(\alpha, \beta_D)$*,* $\tilde{S} \sim \mathrm{Beta}(\alpha_S, \beta_S)$*), a sweep of more than* 250 *parameter cells (demand skews* α, β_D*, capacity factors* $\mathbb{E}[\tilde{S}]$ *up to* 0.8*, the* $C_R \times C_M$ *grid) yields, at* every *large-FCL interior optimum located:* (i) *the floor (13) holds;* (ii) *the uniform-shape benchmark is already negative,* $\Delta_0 < 0$ *(*$\Delta_0^{\max} \approx -1.1$*); and* (iii) *the realized* $\Delta < 0$ *(*$\Delta^{\max} \approx -1.4$*)—the distributional tilt* reinforces *the negative benchmark rather than reversing it. Given the impossibility result, this is the strongest statement the structure admits short of a distribution-specific computation: we therefore state it as the* (numerically established) *resolution, with the residual "for all* F*" universal a conjecture by necessity, not by lack of proof effort. Carbon strictly falls only at the cheap-renewable knife-edge* corner $K_R^* \to \infty$ *(power-law, the window* $(C_R^\circ, C_R^\dagger)$ *of Week-1 Theorem 4(ii)), where* $\mathbb{E}[\mathit{fossil}^*] = 0 < \mathbb{E}[\mathit{fossil}^\circ]$*.*

Remark 3 (Status, and an earlier slip corrected). Week 1 conjectured carbon falls at an interior optimum but never exhibited one. The treatment here is, we believe, as far as the question can be pushed analytically: Lemma 4 proves the structural floor; Proposition 2 proves that the sign is *not* cutoff-decidable (so the conjecture admits no distribution-free proof—a genuine obstruction, not a gap in effort); and Proposition 3 resolves the residual distributional question numerically, finding the benchmark Δ_0 already negative and the tilt reinforcing it at every scanned optimum. Interior carbon reduction is, at best, rare and unwitnessed; the only clean strict fall is the corner. (An earlier draft mis-stated the mechanism as "FCL draws fossil on surplus"; the fossil formula already credits surplus absorption—the correct obstruction is the floor.) The small-FCL regime ($K_M^* < K_R^*$) is outside Role 1's large-FCL mandate and is left to Role 4.

[9] Caveat (as in Theorem 5): the extremal laws attaining $\Delta_{\min}, \Delta_{\max}$ are degenerate and need not be realized by an equilibrium F. The claim is under-determination of sign Δ by capacities and cutoffs over the admissible-conditional-law class, not that an equilibrium attains the extremes.

[10] T3 checks $\Delta \geq \Delta_0$ holds at the scanned optima with the realized (reverse-tilted) densities giving $\Delta \leq \Delta_0$, and that Δ_0 is computed from cutoff+capacity data.

5. Connectedness of the carbon-reduction set

Let $\mathcal{S} := \{C_R \geq 0 : \Delta(C_R) \geq 0\}$ be the set of renewable costs at which the program reduces carbon. Week 1 (Thm. 4) proved $\mathcal{S}$ bounded above for every F; it is non-empty via the cheap-renewable *corner* of the power-law family (Proposition 3: on $(C_R^\circ, C_R^\dagger)$, $K_R^* \to \infty$ gives $\mathbb{E}[\text{fossil}^*] = 0 < \mathbb{E}[\text{fossil}^\circ]$). On the Beta-mixture *interior*, by contrast, $\mathcal{S}$ may be empty (the same proposition). We now give the condition under which $\mathcal{S}$ is a single interval.

Condition 4 (Monotone renewable shadow value, MRSV)**.** At every C_R where the program is carbon-neutral, $\Delta(C_R) = 0$, the saving strictly down-crosses: $\Delta'(C_R) < 0$. Equivalently, at such a C_R the FCL fossil output rises strictly faster in C_R than the no-FCL output, $\frac{d}{dC_R}\mathbb{E}[\text{fossil}^*] > \frac{d}{dC_R}\mathbb{E}[\text{fossil}^\circ]$.

This is a bare sufficient condition; we do *not* have a primitive justification for it. One might guess it follows from a shadow-value ordering $\sigma^* > \sigma^\circ$ ($\sigma := -d\,\mathbb{E}[\text{fossil}]/dK_R$), but that guess is *false*: numerically $\sigma^* < \sigma^\circ$ (the no-FCL renewables carry the larger marginal fossil impact, since they displace fossil directly while the FCL program's renewables partly feed computing). So Condition 4 is genuinely a hypothesis, not a derived fact—and, as the next theorem shows, it is also *not needed* for the family where carbon actually falls.

Theorem 3 (Single-interval carbon-reduction set)**.** (i) *$\mathcal{S}$ is bounded above: there is $B_0 < \infty$ with $C_R > B_0 \Rightarrow K_R^* = K_R^\circ = 0$ and $\Delta(C_R) = -U_0 < 0$, the strictly positive utilization penalty.* (ii) (Power-law/scale family—exact, unconditional.) *$\mathcal{S} = [0, C_R^\dagger]$, a single bounded interval, with the strict-reduction core $\{\Delta > 0\} = (C_R^\circ, C_R^\dagger)$ the open window of Week-1 Theorem 4(ii) (closed-form thresholds $C_R^\circ < C_R^\dagger$). In particular $\mathcal{S} \neq \varnothing$.* (iii) (General F, conditional extension.) *Δ is C^1 in C_R on the large-FCL interior (Corollary 1); under Condition 4, Δ is then single-crossing on $\{C_R : \Delta \geq 0\}$, so any interior reduction set is a single interval. (By Proposition 3 there is no interior reduction in the scanned family, so (iii) is the empty-or-corner case; it matters only if some F outside the scan reduces carbon at an interior optimum.)*

Proof. (i) is Week-1 Theorem 4(i): above the finite renewable-benefit ceiling B_0 the renewable FOC has the corner $K_R^* = K_R^\circ = 0$, leaving only the distribution-free utilization effect, so $\Delta = -U_0 < 0$. (ii) is the Remark below (Week-1 closed forms). For (iii), Δ is C^1 on the large-FCL interior by Corollary 1; we show $\{C_R : \Delta \geq 0\}$ restricted to that region is convex. Suppose not: then there are $C_R^{(1)} < C_R^{(2)} < C_R^{(3)}$ with $\Delta(C_R^{(1)}) \geq 0$, $\Delta(C_R^{(2)}) < 0$, $\Delta(C_R^{(3)}) \geq 0$. By the intermediate value theorem Δ has a zero in $(C_R^{(2)}, C_R^{(3)})$; let z be the smallest such zero. On $[C_R^{(2)}, z)$, $\Delta < 0$ (else there would be an earlier zero), and $\Delta(z) = 0$; hence Δ is non-decreasing as it reaches z from the left, $\Delta'(z) \geq 0$. This contradicts Condition 4, which forces the strict down-crossing $\Delta'(z) < 0$ at every zero z. So no interior dip exists and $\mathcal{S}$ is convex, i.e. an interval; boundedness (i) makes its right endpoint $C_R^\dagger = \sup \mathcal{S}$ finite, and strict down-crossing rules out any second crossing.[11] □

Remark 4 (The power-law set is exactly a single interval, unconditionally). For the family where the actual carbon-reduction set is non-trivial—the power-law/scale family—no condition is needed: by Week-1 (Prop. $C_R^\dagger$), on $(C_R^\circ, C_R^\dagger)$ one has $K_R^* = \infty$, $\mathbb{E}[\text{fossil}^*] = 0$, and K_R° finite with $\mathbb{E}[\text{fossil}^\circ] > 0$, so $\Delta > 0$ *exactly* there, while $\Delta < 0$ for $C_R > C_R^\dagger$ and $\Delta = 0$ for $C_R \leq C_R^\circ$ (both grids hit the unbounded-renewable corner, so both fossil outputs vanish and there is no *strict* reduction). Hence $\mathcal{S} = \{\Delta \geq 0\} = [0, C_R^\dagger]$ is a single bounded interval by the closed-form thresholds, with *no* appeal to Condition 4; the *strict*-reduction set $\{\Delta > 0\} = (C_R^\circ, C_R^\dagger)$ is its open core, and the flat $[0, C_R^\circ]$ is the degenerate "both fully decarbonized" piece. (This is the "carbon strictly falls only in the corner" statement of the abstract: strict fall happens exactly on $(C_R^\circ, C_R^\dagger)$, the corner window.) The

[11] T4: sweeping C_R on the witness family, $\Delta(C_R)$ has exactly one sign change ($+ \to -$), is non-increasing where $\Delta \geq 0$, and is negative at the largest C_R.

MRSV machinery is the general-F extension, and—given Proposition 3, which finds no interior reduction in the scanned family—it is the only place a general-F reduction set could be non-trivial off the corner. The C^1-ness that (iii) needs is now *proved* on the large-FCL interior (Corollary 1), so only Condition 4 itself is a hypothesis, and it is close to its own conclusion ("strictly down-crosses at every zero" *is* single-crossing). We make no claim that it holds: as noted above, the natural shadow-value justification fails ($\sigma^* < \sigma^\circ$, verified in §7). This costs the deliverable nothing, because part (ii)—the only family where carbon actually falls—gives the single interval *exactly*, needing neither Condition 4 nor C^1-ness. Part (iii) is included only to record what a general-F extension would require, and is vacuous in the scanned family (no interior reduction, Proposition 3).

6. Sharpness: the necessary-and-sufficient channel condition and tight bounds

6.1. The sharp channel condition is a path integral.

Week 1 used pointwise-monotone g as a *sufficient* condition for $K_R^* > K_R^\circ$. The sharp condition is integral, not pointwise.

Theorem 4 (Necessary and sufficient channel condition)**.** *On the open large-FCL interior interval where Condition 1 holds ($\mathcal{H} \prec 0$), write $\lambda(C_M) := -dK_R^*/dC_M = \kappa(C_M)\, f(K_F^*) f(a)\,[\,g(a) - g(K_F^*)\,]$ with $\kappa(C_M) = (\beta + p - R)(R - c)/|\det \mathcal{H}| > 0$. For any compact subinterval $[C_M, C_M']$ of that open interval,*

$$K_R^*(C_M) - K_R^*(C_M') = \int_{C_M}^{C_M'} \lambda(t)\, dt, \tag{15}$$

and, letting $C_M' \uparrow \bar{C}_M$ where $K_R^*(C_M') \to K_R^\circ$ by continuity (Week 1), $K_R^*(C_M) > K_R^\circ$ if and only if *$\int_{C_M}^{\bar{C}_M} \lambda > 0$. Pointwise $g(a) \geq g(K_F^*)$ (Condition 2) makes $\lambda \geq 0$ and is therefore* sufficient*; it is* not necessary*, since the integral can be positive while the integrand is negative on part of the path.*

Proof. On the open interior interval Condition 1 gives $\mathcal{H} \prec 0$, so by the implicit function theorem $C_M \mapsto (K_R^*, K_F^*, K_M^*)$ is C^1 there, and $\lambda = -dK_R^*/dC_M$ is *continuous*. On a compact subinterval $[C_M, C_M']$ a continuous derivative is integrable, so K_R^* is C^1 (a fortiori absolutely continuous) and the fundamental theorem of calculus gives (15) exactly. The factorization of λ is the Week-1 cofactor identity divided by $|\det \mathcal{H}| > 0$. As $C_M' \uparrow \bar{C}_M$ the first FCL unit's marginal profit vanishes, $K_M^* \to 0$, and the system collapses to the no-FCL optimum, $K_R^*(C_M') \to K_R^\circ$ (Week 1, continuity at the adoption boundary). We *define* the improper integral $\int_{C_M}^{\bar{C}_M} \lambda$ as $\lim_{C_M' \uparrow \bar{C}_M} \int_{C_M}^{C_M'} \lambda$; by (15) this equals $\lim_{C_M' \uparrow \bar{C}_M} [K_R^*(C_M) - K_R^*(C_M')] = K_R^*(C_M) - K_R^\circ$, a finite limit—so the improper integral converges regardless of whether λ stays bounded as $C_M' \uparrow \bar{C}_M$ (where $\det \mathcal{H} \to 0$): convergence is inherited from the antiderivative's finite limit, not from integrability of λ. Positivity of $K_R^*(C_M) - K_R^\circ$ is then equivalent to positivity of $\int_{C_M}^{\bar{C}_M} \lambda$. Sufficiency of Condition 2 is $\lambda \geq 0$ pointwise; non-necessity is the elementary fact that an integral of a sign-changing integrand can be positive.[12] □

6.2. The two-sided bounds are tight.

Theorem 5 (Tight magnitude bounds)**.** *Fix the equilibrium capacities and q°, and range over all conditional laws consistent with the distribution-free cutoffs (these laws need not all be realized by an equilibrium F; the extremizers below are degenerate, not equilibrium laws). With Δ written as the affine map (10) of the conditional means $\mu_{\text{low}}^* \in (-K_R^*, a)$, $\mu_{\text{mid}}^\circ \in (0, K_F^\circ)$,*

$$\Delta_{\min} := K_F^\circ(1 - \varphi^\circ) - K_F^* \;\leq\; \Delta \;\leq\; K_F^\circ - K_F^* + \tfrac{C_M}{R-c}(a + K_R^*) := \Delta_{\max}, \tag{16}$$

[12] T5: on the interior band $\int_{C_M}^{C_M'} (-dK_R^*/dC_M)\, dt$ matches $K_R^*(C_M) - K_R^*(C_M')$ to $\sim 2\%$ (4.21 vs 4.13); the sufficient-not-necessary logic is shown by a sign-mixed integrand with positive integral. A *realized* F with $\lambda < 0$ on a sub-path is Role 3's reversed-channel counterexample, outside this deliverable.

and both bounds are the exact sup/inf of the affine map, attained in the limit: $\Delta_{\min}$ *as the* FCL *conditional law concentrates at the upper endpoint (*$\mu^*_{\text{low}} \uparrow a$*) and the no-*FCL *law at the lower endpoint (*$\mu^\circ_{\text{mid}} \downarrow 0$*);* $\Delta_{\max}$ *as the laws concentrate oppositely (*$\mu^*_{\text{low}} \downarrow -K^*_R$*,* $\mu^\circ_{\text{mid}} \uparrow K^\circ_F$*,* $q^\circ \downarrow 0$*).*

Proof. Δ is affine and monotone in each conditional mean: $\partial\Delta/\partial\mu^*_{\text{low}} = -C_M/(R-c) < 0$ and $\partial\Delta/\partial\mu^\circ_{\text{mid}} = \varphi^\circ - q^\circ > 0$. A conditional mean ranges over the open interval between its endpoints, with the endpoints approached by laws concentrating there (e.g. a two-point law placing mass $\to 1$ at the endpoint while preserving the cutoff probability). Plugging the extreme means into (10) gives (16), and the attaining sequences are the concentrating laws just named; the bounds are therefore the exact sup/inf, i.e. tight.[13] □

7. Numerical validation

All identities are checked in `verify_week2.py` against adaptive quadrature of the profit integral (1), on non-power-law Beta scale mixtures $\tilde{x} = \tilde{D} - K_R\tilde{S}$. T1 confirms the recursive triangle (interiority-only (K_F, K_M)-block, scalar Schur, $\mathcal{H} \prec 0$, unique Ψ zero); T2 the uniform endpoint inequality along the homotopy; T3 the exact endogenous identity, the sufficient-bound algebra, and the Proposition 3 scan ($\Delta < 0$ at every large-FCL interior optimum, $\Delta^{\max} \approx -1.4$); T4 the corner non-emptiness $C^\circ_R < C^\dagger_R$, the large-cost anchor, and the single interval; T5 the FTC/path-integral identity and the tight bounds $\Delta_{\min} \le \Delta \le \Delta_{\max}$ attained at the endpoints. The suite reports a single pass count.

[13] T5b: $\Delta(\mu^*_{\text{low}}{=}a, \mu^\circ_{\text{mid}}{=}0) = \Delta_{\min}$ and $\Delta(-K^*_R, K^\circ_F) = \Delta_{\max}$ exactly, and the realized Δ lies strictly inside.